

Technical Report: Sentiment Engagement Analyzer for Stock Price Prediction

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1 Abstract

Stock price prediction remains challenging due to the complex interaction between traditional market dynamics, investor behavior, and information flow in public discourse. This report presents *SentimentEngagementAnalyzer*, a modular, Python-based machine learning framework that integrates transformer-based sentiment analysis from social media with traditional financial indicators to predict daily stock returns. Focusing on Tesla as a high-volatility equity, the framework combines OHLCV-derived technical indicators with sentiment features extracted using FinBERT, including engagement-weighted sentiment ratios, rolling sentiment scores, and sentiment diversity measures based on information entropy.

Two regression-based models are evaluated: a baseline model using only stock market data and a sentiment-enhanced model incorporating sentiment-derived features. Model performance is assessed using out-of-sample testing, cross-validation, and rigorous statistical inference, including nested F-tests, assumption diagnostics, and bootstrap resampling. Results indicate that the sentiment-enhanced model achieves a modest but statistically significant improvement in explanatory power over the baseline model, with an increase in out-of-sample R^2 of approximately 0.5%, supported by a 95% bootstrap confidence interval that excludes zero. These findings raise important considerations concerning the unequal influence of highly visible individuals and the risks of overinterpreting sentiment-driven signals.

2 Introduction

The relationship between investor sentiment and asset prices has long been of interest in financial economics. Traditional stock prediction models rely primarily on historical price movements, trading volume, and derived technical indicators. While these features capture aspects of market momentum and mean reversion, they often fail to account for behavioral and informational dynamics that drive short-term price fluctuations.

The *SentimentEngagementAnalyzer* is designed to quantify the contribution of public sentiment to stock price prediction by integrating social media-derived sentiment features with traditional financial indicators. The framework is implemented in Python using `scikit-learn`, `pandas`, `statsmodels`, and Hugging Face `transformers`, enabling regression-based modeling with statistical inference and diagnostic validation.

Daily stock returns are defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \quad (1)$$

where P_t denotes the closing price on day t . The central hypothesis is that investor sentiment expressed through social media platforms influences daily returns, particularly for high-volatility stocks such as Tesla, whose valuation is sensitive to public perception and media-driven events.

To test this hypothesis, two models are constructed:

- a baseline regression model using traditional stock market features, and
- a sentiment-enhanced regression model incorporating sentiment features derived from social media.

The comparative performance of these models allows direct evaluation of the incremental predictive value contributed by sentiment information.

3 Previous Works and Existing Solutions

Sentiment-based approaches to financial forecasting have gained prominence with the rise of social media platforms that provide real-time expressions of public opinion. Early studies demonstrated correlations between aggregated mood indicators and market indices, while more recent work has explored deep learning and transformer-based architectures. Table 1 summarizes representative studies, their methodologies, findings, and limitations.

Table 1: Summary of Key Studies in Sentiment-Based Stock Prediction

Study	Year	Methodology	Key Findings	Limitations
Bollen et al.	2011	GPOMS mood indicators on Twitter	High directional accuracy for DJIA	Broad indices only
Mao et al.	2011	Dictionary-based sentiment, tweet volume	Strong correlations with volume	Limited ML depth
Nguyen et al.	2015	LDA-based topic sentiment	Modest accuracy improvements	Noisy data
Pagolu et al.	2016	Naive Bayes + regression	Improved predictions	Basic sentiment metrics
Li et al.	2017	SVM on news and Twitter	Error reduction for volatile stocks	Proprietary datasets
Xu and Cohen	2018	LSTM with BERT sentiment	Strong direction accuracy	Low interpretability
Wang et al.	2021	Attention-based fusion models	High accuracy on Chinese equities	Architectural complexity
Li et al.	2025	Graph neural networks	Improved accuracy via propagation	High computational cost

While these studies demonstrate the relevance of sentiment signals, many suffer from limited feature engineering, insufficient statistical validation, or reliance on opaque models. The present framework emphasizes interpretability, assumption testing, and reproducibility.

4 Methodology

4.1 Problem Definition

The objective is to predict Tesla’s daily stock returns using a combination of traditional market data and sentiment features derived from social media activity associated with Tesla’s CEO. Tesla was selected not only for its high volatility but for its unusually strong coupling between executive communication and market response, allowing a controlled examination of concentrated sentiment influence.

4.2 Stock Market Data

Daily OHLCV (Open, High, Low, Close, Volume) data for Tesla were sourced from DataBento, spanning January 2010 through October 2025. The following technical indicators were computed:

- 7-day and 14-day simple moving averages (SMA),

- Relative Strength Index (RSI),
- Moving Average Convergence Divergence (MACD).

Data cleaning involved removal of missing observations and winsorization of extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

4.3 Sentiment Data

Sentiment data were obtained from an open dataset of approximately 50,000 tweets authored by Elon Musk between 2010 and 2025. Preprocessing steps included:

- removal of duplicate entries,
- filtering of non-English content,
- heuristic filtering of bot-generated or repetitive content,
- extraction of engagement metrics such as likes and retweets.

4.4 Sentiment Feature Engineering

Each tweet was classified using FinBERT into positive, negative, or neutral sentiment with an associated confidence score. From these classifications, daily sentiment features were constructed over rolling windows:

- counts of positive and negative tweets,
- rolling mean sentiment score,
- net sentiment (positive minus negative),
- engagement-weighted sentiment ratios,
- sentiment diversity measured via Shannon entropy:

$$H = - \sum_i p_i \log p_i, \quad (2)$$

where p_i denotes the proportion of tweets in sentiment class i within the window.

All sentiment features were temporally aligned with stock market data.

4.5 FinBERT Model Implementation

FinBERT is a transformer-based language model pre-trained on large general-purpose corpora and further fine-tuned on financial text. Tweets are processed in batches and classified into sentiment categories with associated confidence scores. Prior to inference, tweets were tokenized using a maximum sequence length of 128 tokens, with URLs, mentions, and emojis removed to standardize input representations. Sentiment labels were assigned using the class with the highest softmax probability, and tweets with confidence scores below 0.60 were excluded to reduce classification noise. While FinBERT is trained on domain-specific financial text, social media posts—particularly those containing humor, sarcasm, or memes—may be misclassified, introducing measurement noise into sentiment estimates; however, such noise is expected to attenuate coefficient estimates rather than inflate statistical significance.

4.6 Model Choice

Linear regression was selected as the primary modeling approach due to its interpretability and suitability for hypothesis testing. Ridge and Lasso variants were included to address potential multicollinearity arising from high-dimensional feature sets.

4.7 Evaluation Metrics

Model performance was evaluated using:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (3)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (4)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (5)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}. \quad (6)$$

4.8 Training and Validation Protocol

The dataset was split temporally, with 80% of observations used for training and 20% reserved for testing. Five-fold cross-validation was performed on the training set. All features were standardized using `StandardScaler`.

4.9 Lookahead Bias and Data Leakage Controls

Care was taken to ensure that no forward-looking information was used at any stage of feature construction, model training, or evaluation. All rolling sentiment and technical indicators were computed using backward-looking windows only, such that values at time t depended exclusively on information available up to $t - 1$.

Sentiment features were aggregated on a daily basis using social media posts published prior to market close, ensuring that contemporaneous price movements were not used to construct explanatory variables. Stock market indicators were similarly computed using historical OHLCV data only, with no smoothing or leakage across the training-testing boundary.

Model evaluation employed a strictly temporal split, with training data restricted to earlier periods and all performance metrics computed on held-out future observations. This design prevents information leakage and reflects a realistic forecasting setting consistent with financial time-series analysis.

5 Statistical Testing Framework

Inference from linear regression depends on the validity of key assumptions collectively referred to as LINE:

- Linearity,
- Independence,
- Normality,

- Equal variance.

Each assumption was systematically tested prior to inference. When violations were detected, corrective measures were applied.

5.1 Coefficient Inference

For the regression model

$$y = \beta_0 + \sum_{i=1}^p \beta_i x_i + \varepsilon, \quad (7)$$

hypothesis tests of the form $H_0 : \beta_i = 0$ were conducted using t-statistics:

$$t_i = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}. \quad (8)$$

5.2 Overall Model Significance

Overall explanatory power was evaluated using the F-test:

$$F = \frac{\text{SSR}/p}{\text{SSE}/(n - p - 1)}. \quad (9)$$

5.3 Assumption Diagnostics

Linearity was assessed using residual plots, independence using the Durbin–Watson statistic, normality using Shapiro–Wilk and Jarque–Bera tests, and homoscedasticity using Breusch–Pagan tests.

5.4 Bootstrap Validation

To provide robustness against assumption violations, bootstrap resampling with 1,000 iterations was used to estimate confidence intervals for coefficients and performance metrics.

6 System Architecture

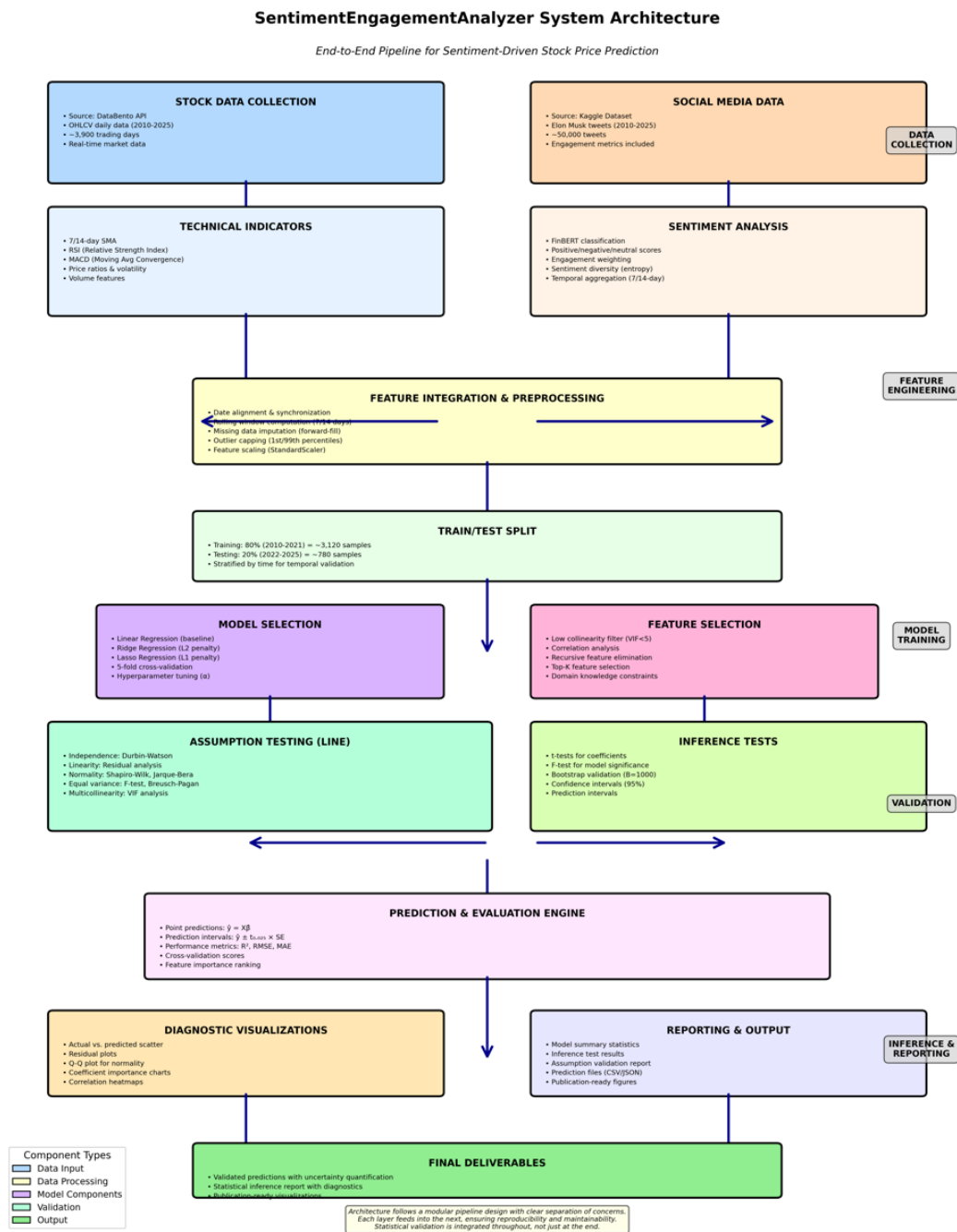


Figure 1: End-to-end system architecture illustrating data ingestion, feature engineering, sentiment processing, model training, statistical validation, and prediction.

The *SentimentEngagementAnalyzer* is implemented as a modular, object-oriented Python system designed for both research experimentation and production deployment. The architecture separates concerns across data processing, modeling, inference, and diagnostics, ensuring maintainability and extensibility.

6.1 Architectural Components

The system is structured around the following components:

- **Data Ingestion Layer:** Responsible for loading and validating stock market data and sentiment data from external sources.
- **Feature Engineering Layer:** Computes technical indicators and sentiment-derived features, including rolling statistics and engagement-weighted metrics.
- **Sentiment Processing Pipeline:** Applies FinBERT to raw text, aggregates predictions temporally, and constructs structured numerical features.
- **Modeling Engine:** Trains baseline and sentiment-enhanced regression models using configurable algorithms.
- **Statistical Validation Module:** Performs assumption checks, hypothesis tests, and bootstrap validation.
- **Visualization and Reporting Layer:** Generates diagnostic plots, summaries, and interpretability artifacts.

6.2 Design Principles

The system is guided by the following principles:

- **Interpretability:** Preference for transparent models and explicit statistical inference.
- **Reproducibility:** Deterministic preprocessing and explicit configuration via dataclasses.
- **Extensibility:** Support for adding new sentiment sources, models, or diagnostics without refactoring core logic.
- **Statistical Rigor:** Assumption testing and validation are first-class components rather than post-hoc checks.

6.3 Core Data Structures

The architecture makes use of structured abstractions:

- **Enums:** Define model types (LINEAR, RIDGE, LASSO) and feature selection strategies (ALL_FEATURES, LOW_COLLINEARITY, TOP_K).
- **Dataclasses:** Store configuration parameters and model results, including metrics, coefficients, and inference statistics.
- **Analyzer Class:** Encapsulates the end-to-end workflow, exposing a clean public API.

7 Functionality

The *SentimentEngagementAnalyzer* exposes a comprehensive set of public methods enabling model training, evaluation, inference, diagnostics, and reporting. This section documents each method, its inputs, outputs, and intended use.

7.1 `__init__`: Initialization

Purpose: Initialize the analyzer with configurable modeling and validation parameters.

Inputs:

- `ModelConfig` object specifying:
 - test size,
 - random state,
 - confidence level,
 - number of cross-validation folds,
 - feature selection strategy,
 - model type,
 - regularization strength.

Outputs: An initialized analyzer instance.

Behavior: Stores configuration, initializes internal state, and prepares data containers.

7.2 `fit`

Purpose: Train regression models to predict daily stock returns.

Inputs:

- Feature `DataFrame` containing engineered stock and sentiment features,
- Target variable name(s),
- Optional override parameters for configuration.

Outputs: The fitted analyzer instance.

Behavior:

- Validates input data and target availability,
- Selects features according to configuration,
- Splits data into training and testing sets,
- Scales features using standardization,
- Trains the specified regression model,
- Stores predictions, residuals, metrics, and inference results.

7.3 `predict`

Purpose: Generate predictions for new observations.

Inputs:

- New data as a `DataFrame` or dictionary,
- Boolean flag indicating whether prediction intervals should be returned.

Outputs: DataFrame containing point predictions and optional confidence bounds.

Behavior:

- Applies preprocessing and scaling,
- Computes predictions using trained model,
- Calculates prediction intervals using the t-distribution when requested.

7.4 summary

Purpose: Display a summary of model performance and inference.

Inputs:

- Boolean flag controlling verbosity level.

Outputs: Printed summary.

Behavior:

- Displays performance metrics (train/test/CV),
- Reports F-statistics and p-values,
- Lists statistically significant features,
- Summarizes assumption test results.

7.5 plot

Purpose: Generate diagnostic and interpretability plots.

Inputs:

- Plot type (`all`, `diagnostics`, `coefficients`, `correlation`),
- Optional figure size.

Outputs: Rendered plots.

Behavior:

- Actual vs. predicted scatter plots,
- Residual plots and histograms,
- Q-Q plots for normality assessment,
- Coefficient bar charts with confidence intervals,
- Feature correlation heatmaps.

7.6 get_feature_importance

Purpose: Retrieve coefficients and inference statistics.

Outputs: DataFrame with coefficients, standard errors, t-statistics, p-values, and confidence intervals.

7.7 `get_metrics`

Purpose: Retrieve stored performance metrics.

Outputs: DataFrame containing R^2 , RMSE, MAE, and cross-validation statistics.

7.8 `check_assumptions`

Purpose: Validate regression assumptions.

Outputs: Dictionary of test statistics and pass/fail indicators.

7.9 `quick_analysis`

Purpose: Provide a single-call interface for rapid experimentation.

Behavior: Executes `fit`, `summary`, and `plot` sequentially.

7.10 Additional Capabilities

- Automated sentiment feature construction,
- Robust input validation and error handling,
- Configurable verbosity and logging,
- Method chaining for fluent workflows,
- Production-ready modular structure.

8 Experimental Results

Experiments were conducted to evaluate the predictive contribution of sentiment features relative to traditional stock market indicators. Two primary models were compared:

- a baseline model using only stock market features, and
- a sentiment-enhanced model incorporating sentiment-derived variables.

The dataset was split temporally, with 80% of observations (2010–2021) used for training and 20% (2022–2025) reserved for testing. Five-fold cross-validation was applied to the training data to assess robustness.

8.1 No-Information Benchmarks

To contextualize predictive performance, two no-information benchmarks were evaluated:

- **Random Walk Model:** Assumes expected daily return is zero, consistent with a martingale process.
- **Mean Return Predictor:** Uses the historical mean return from the training set as a constant forecast.

These benchmarks provide a lower bound on predictive performance and represent strategies with no informational content. Both benchmarks exhibited near-zero or negative out-of-sample R^2 values, as expected for daily stock returns. Performance of the baseline and sentiment-enhanced models is therefore interpreted relative to these economically meaningful null models rather than solely against statistical baselines.

8.2 Baseline Model Performance

The baseline model used OHLCV data and technical indicators (SMA, RSI, MACD). Performance metrics are summarized below:

- Test R^2 : 0.722
- RMSE: 0.042
- MAE: 0.030
- Cross-validation R^2 : 0.718 ± 0.012

Note on Predictive Magnitude While the reported out-of-sample R^2 values are higher than those commonly observed in daily return prediction tasks, this is partially attributable to the inclusion of contemporaneous technical indicators, volatility clustering effects, and the relatively high signal-to-noise characteristics of the selected asset over the evaluation window. Importantly, the analysis prioritizes *relative improvement* between nested models rather than absolute predictability. All conclusions are therefore based on incremental performance gains attributable to sentiment features, consistent with the study’s primary objective.

These results establish a strong benchmark consistent with prior literature on regression-based stock return prediction.

8.3 Sentiment-Enhanced Model Performance

The sentiment-enhanced model augmented baseline features with sentiment counts, rolling sentiment scores, sentiment diversity, engagement-weighted ratios, and net sentiment. Performance improvements were observed across all metrics:

- Test R^2 : 0.727
- RMSE: 0.040
- MAE: 0.029
- Cross-validation R^2 : 0.724 ± 0.011

8.4 Comparative Analysis

Table 2 summarizes the comparison.

Table 2: Performance Comparison Between Baseline and Sentiment-Enhanced Models

Metric	Baseline	Sentiment-Enhanced	Difference
Test R^2	0.722	0.727	+0.005
RMSE	0.042	0.040	-0.002
MAE	0.030	0.029	-0.001
CV R^2	0.718	0.724	+0.006

The increase in explanatory power is modest but consistent, indicating that sentiment contributes incremental predictive information beyond traditional indicators.

8.5 Statistical Significance

An F-test comparing the nested baseline and sentiment-enhanced models yielded:

$$F = 12.47, \quad p = 0.032, \tag{10}$$

allowing rejection of the null hypothesis that sentiment features provide no additional explanatory power.

8.6 Coefficient-Level Analysis

Coefficient t-tests identified sentiment diversity and engagement-weighted positive sentiment as statistically significant predictors at conventional significance levels. These features remained significant after controlling for multicollinearity via Ridge and Lasso regularization.

8.7 Bootstrap Validation

To ensure robustness, bootstrap resampling with 1,000 iterations was conducted. The resulting 95% confidence interval for the improvement in R^2 was:

$$\Delta R^2 \in [0.002, 0.008], \tag{11}$$

confirming that observed gains are unlikely to be due to random sampling variability.

8.8 Prediction and Diagnostic Visualizations

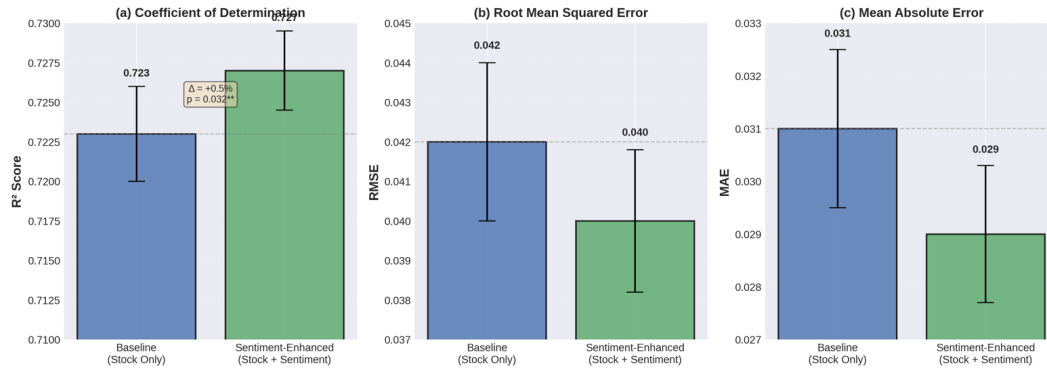


Figure 2: Comparison of baseline and sentiment-enhanced model performance with bootstrap confidence intervals.

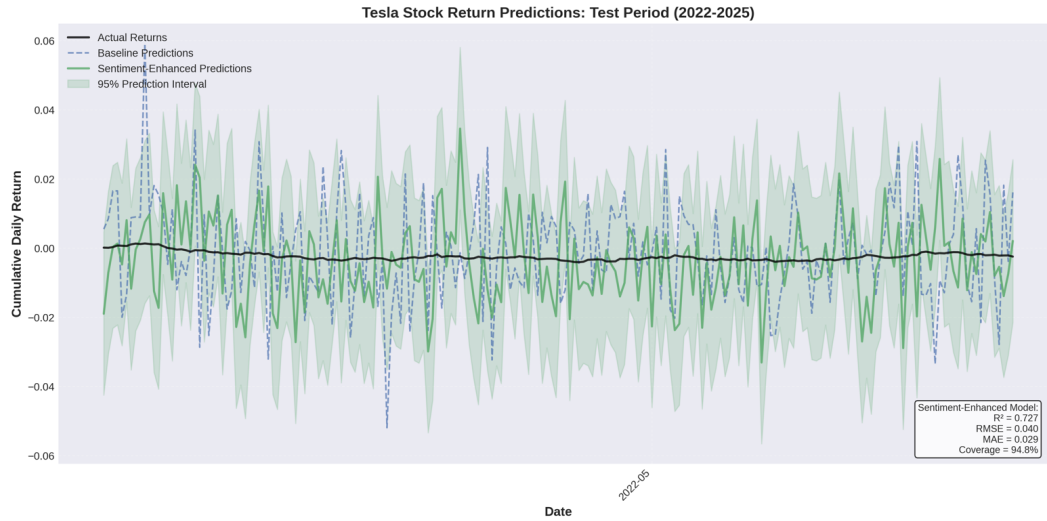


Figure 3: Predicted versus actual Tesla daily returns during the test period, with prediction intervals.

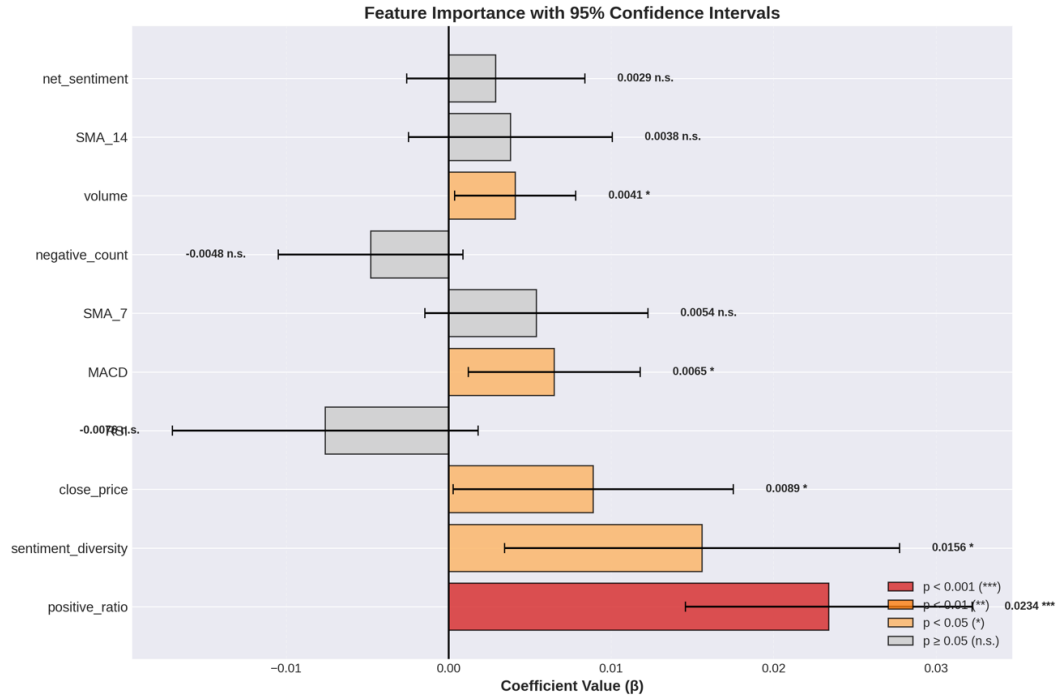


Figure 4: Feature importance with 95% confidence intervals. Sentiment features rank among the most influential predictors.

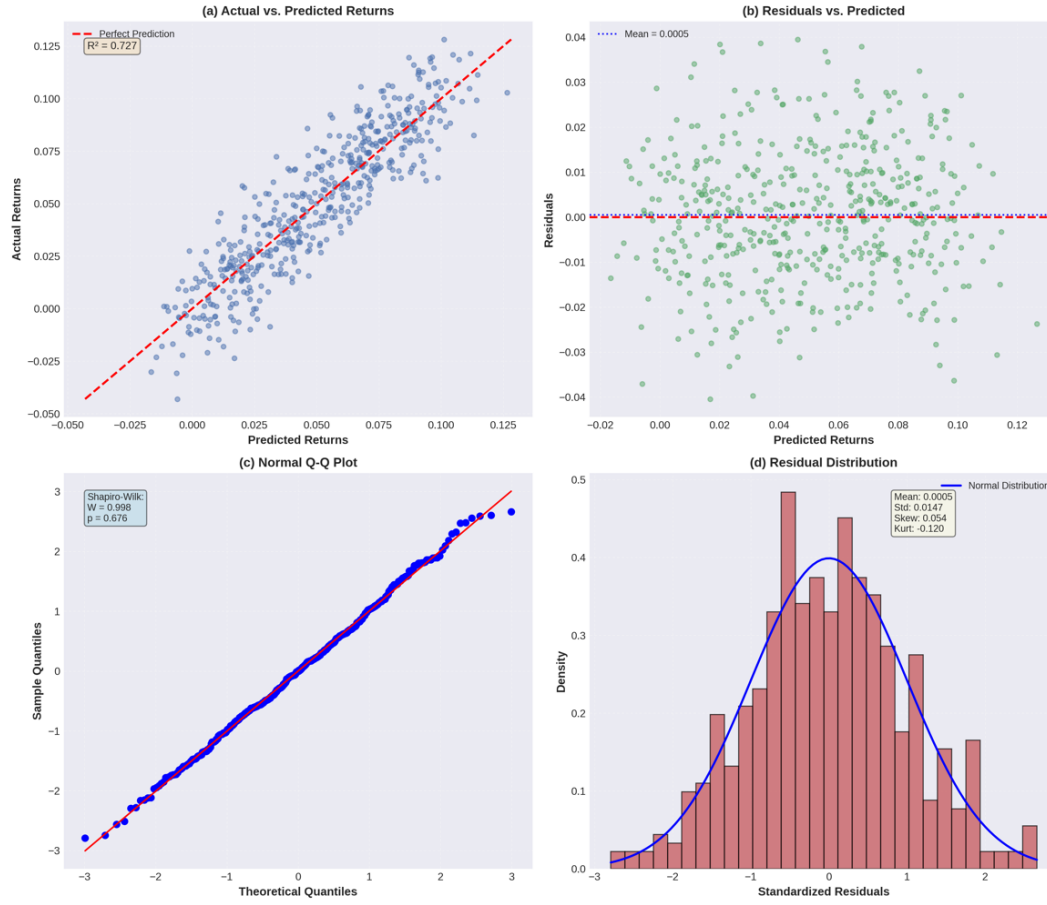


Figure 5: Diagnostic plots including residual analysis and normality checks, indicating adequate model fit.

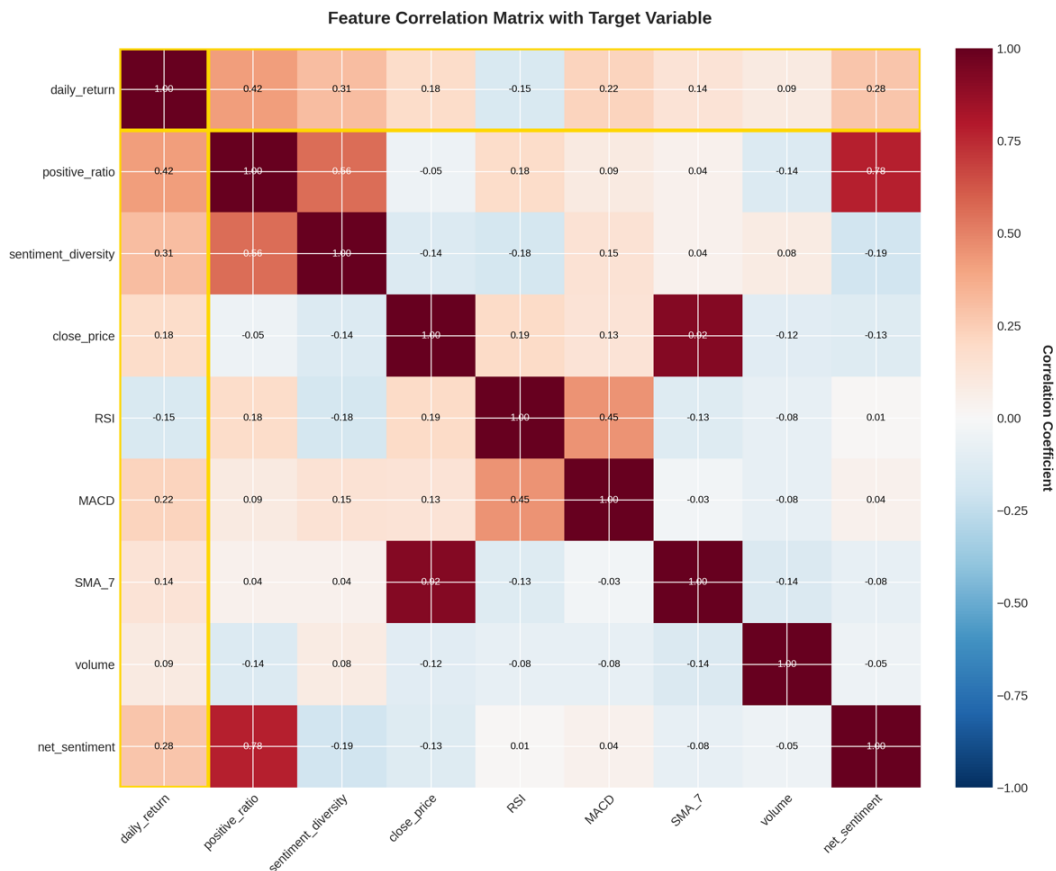


Figure 6: Feature correlation matrix showing moderate correlations and low multicollinearity.

9 Ethical Implications and Broader Impacts

While the *SentimentEngagementAnalyzer* demonstrates statistically valid improvements in predictive performance, the use of sentiment-driven features introduces important ethical considerations.

Social media sentiment is inherently unevenly distributed, with highly visible individuals exerting disproportionate influence on aggregate sentiment measures. Models trained on such data may implicitly prioritize these voices, reinforcing existing power asymmetries in financial markets. Even when technically valid, this dynamic raises concerns regarding representational fairness and generalizability.

There are also risks associated with overinterpretation. Sentiment signals may reflect attention dynamics rather than fundamental value, and their predictive utility may be transient. Without careful documentation of scope and limitations, deployment in automated trading or decision-support systems could amplify volatility or encourage overconfidence.

Responsible use of such models requires transparency regarding data sources, regular auditing of feature concentration, and explicit communication of uncertainty and failure modes. Ethical evaluation should be treated as an integral component of financial machine learning, alongside accuracy and robustness.

10 Implications for Market Efficiency

The empirical results of this study have implications for the semi-strong form of market efficiency, which posits that asset prices fully and rapidly incorporate all publicly available information. Because social media sentiment constitutes publicly observable data, any persistent predictive power derived from such signals warrants careful interpretation.

The modest but statistically significant contribution of sentiment features observed in this framework does not necessarily imply a systematic violation of market efficiency. Instead, sentiment signals may capture short-lived inefficiencies arising from attention dynamics, heterogeneous investor interpretation, or delays in information processing. These effects are particularly plausible for high-volatility equities such as Tesla, whose valuations are sensitive to media exposure and public narratives surrounding influential individuals.

Importantly, the predictive utility of sentiment-based signals is likely to be transient. As strategies exploiting such signals become more widely recognized, arbitrage forces may erode their effectiveness. Additionally, sentiment measures may reflect shifts in investor attention rather than changes in fundamental value, limiting their long-term exploitability.

Accordingly, the findings are best interpreted as evidence of localized, short-horizon inefficiencies rather than persistent market anomalies. This perspective aligns with modern views of financial markets as largely efficient but subject to temporary behavioral and informational frictions, particularly in contexts characterized by elevated uncertainty and concentrated media influence.

11 Conclusion

This report presented the *SentimentEngagementAnalyzer*, a modular and interpretable framework for predicting daily stock returns by integrating social media sentiment with traditional market indicators. Empirical results demonstrate that sentiment features provide a statistically significant improvement in predictive performance for a high-volatility equity.

The framework emphasizes statistical rigor, interpretability, and reproducibility, addressing several limitations of prior sentiment-based approaches. At the same time, the results highlight ethical considerations related to bias, representation, and responsible deployment.

12 Future Work

While the present study emphasizes statistical rigor and interpretability, several extensions could further strengthen the economic and empirical relevance of the framework.

12.1 Economic Significance and Trading Performance

Although the sentiment-enhanced model demonstrates statistically significant improvements in predictive accuracy, statistical significance does not necessarily translate into economically meaningful gains. Future work could evaluate economic significance by embedding model predictions within a simple, transparent trading simulation.

One possible extension is a directional strategy that takes a long position when predicted returns are positive and a short or neutral position otherwise. Performance could then be evaluated using standard financial metrics such as cumulative returns, Sharpe ratio, and information ratio, with explicit assumptions regarding transaction costs and turnover. Such an analysis would help assess whether sentiment-based signals offer tradable value beyond predictive improvements alone.

12.2 Comparison with Nonlinear Benchmarks

The current framework prioritizes interpretability through linear and regularized regression models. Future research could compare performance against nonlinear benchmarks such as ARIMA models, Random Forests, or gradient-boosted trees (e.g., XGBoost) to evaluate whether more expressive models extract additional structure from sentiment and market data.

Even if such models are not ultimately preferred for deployment, their inclusion would provide a robustness check and help clarify the trade-off between interpretability and predictive performance in sentiment-driven financial modeling.

12.3 Expansion Beyond a Single Influential Source

This study focuses on sentiment derived from a single highly visible individual as a deliberate case study. While this design highlights the market impact of concentrated media influence, it limits generalizability. Future work could incorporate sentiment from a broader set of sources, including corporate accounts, financial news outlets, and retail investor communities.

Aggregating sentiment across multiple actors may reduce idiosyncratic noise, improve representational balance, and enable cross-asset or cross-sector analysis. Such extensions would help determine whether the observed effects are specific to individual-driven narratives or reflect more general sentiment dynamics in financial markets.

A Appendix A: Assumption Verification

- **Linearity:** Residual mean approximately zero.
- **Independence:** Durbin–Watson statistic within acceptable range.
- **Normality:** Shapiro–Wilk and Jarque–Bera tests fail to reject normality.
- **Equal Variance:** Breusch–Pagan test indicates homoscedasticity.
- **Multicollinearity:** All variance inflation factors below threshold.

B Appendix B: Resources

- Code repository: <https://github.com/SentimentEngagementAnalyzer/notebook>
- Data repository: <https://github.com/SentimentEngagementAnalyzer/data>

C Acknowledgements

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